MIL-S-28618B(YD)
5 March 1982
SUPERSEDING
MIL-S-28618A(YD)
22 April 1977

MILITARY SPECIFICATION

SCRAPER, EARTHMOVING, 21 CUBIC YARD

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a four wheel drive scraper having a minimum bowl capacity of 21 cubic yards (cu yd), powered by two diesel engines. The scraper is intended for general earthmoving work.

2. APPLICABLE DOCUMENTS

2.1 *Government documents.*

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-3351 - Tractor, Full Tracked, Low Speed, Tractor, Wheeled, Agriculture and Tractor, Wheeled, Industrial, and Their Attachments, Packaging of.

STANDARDS

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3805

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (DoL)

Occupational Safety and Health Act (OSHA) with Supplemental Revisions.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The scraper shall consist of two power plants, two power trains, a scraper bowl, and a chassis mounted on four drive wheels.

3.2 First article. When specified (see 6.1), the contractor shall furnish a scraper for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard commercial product. The scraper shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which, has been sold or is being currently offered for sale, on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

* 3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.5 Safety. All rotating or reciprocating parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded. The scraper shall be equipped with a locking device to prevent scraper articulation during lifting operation. The scraper shall comply with Occupational Safety and Health Administration regulations in effect at time of manufacture. Scraper shall be provided with at least the following safety equipment:

a. Windshield with power-operated windshield wiper(s).
b. Horn.
c. Backup alarm device conforming to SAE J994.
d. Seat belt for operator's seat.
e. Rollover protective structure at operator's station.
f. Fenders over all wheels.
g. Scraper cushioning device, located in the gooseneck area joining the front and rear portions of the scraper or located at the axle as a suspension system.

3.6 Identical items. All units of the same classification furnished under and specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviations will be acceptable without written approval of the contracting officer.

3.7 Dimensions, weights, and capacities. Scraper shall conform to the dimensions, weights, and capacities specified in table I.

*TABLE I. Dimensions, weights, and capacities.

	Minimum	Maximum
Bowl dimensions and capacity:		
Yardage ratings - SAE J741		
Struck capacity	21 cu yd	
Heaped capacity	28 cu yd	
Payload	75,000 lb	
Cutting depth	12 in.	
Cutting width	10 ft.	
Spreading depth	16 in.	
Apron opening - bowl 6 in. off ground	6 ft.	
Nonstop turning width		45 ft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

3.8 Performance. Scraper shall withstand the hard usage encountered without permanent deformation or damage, while performing the following operations:

a. Load undisturbed or recompacted soil (see 6.4.1), at a minimum rate of not less than 9 cu yd per minute without assistance from auxiliary equipment.
b. With a heaped payload travel at an average speed of not less than 25 miles per hour (mph) on maintained earth haul roads (see 6.4.2).
c. With a heaped payload negotiate a grade having a minimum slope of not less than 20 percent.
d. Without payload traverse a longitudinal grade having a minimum slope of not less than 30 percent.
e. Dump and spread the soil without assistance from auxiliary equipment.

The scraper in general, and the electrical, fuel, hydraulic, and exhaust systems in particular, shall be designed to resist harmful effect of dust, water, and temperature. Scraper shall be capable of satisfactorily performing the operations specified herein in any ambient temperature from -20° Fahrenheit (F) to +105°F.

3.9 Ease of maintenance. Design of the scraper shall permit ready accessibility to all parts for maintenance and repair in the field by use of conventional general-purpose tools associated with these scrapers. Replacement and adjustment of component parts and accessories shall be accomplished with minimum drainage requirements and disturbance to other elements of the scraper. To assure ease of maintenance provisions, to facilitate maintenance, adjustment and inspection of the scraper, access openings shall conform to the minimum dimensions of SAE J925.

3.10 Chassis. The chassis shall consist essentially of the tractor's main frame, scraper's draft frame, and push frame. The frames shall be constructed of structural steel sections or steel plates with members joined by using one of the arc welding processes. To withstand the stresses imposed, and to prevent permanent deformation or damage to the scraper when loaded to gross vehicle weight (gvw) (see 6.4.3), the frames shall be braced and reinforced wherever necessary.

3.10.1 Tractor frame. The tractor frame shall support or incorporate as integral parts the bumper, towing hook(s), crankcase guard, fenders, oscillating kingpin hitch assembly, operator's station, power plant, and power train. Frame shall be mounted on two power driven pneumatic tired wheels. Each towing hook shall withstand a pulling force equivalent to not less than 150 percent of the scraper's gvw. Minimum throat opening of each hook shall be not less than 1-1/2 inches. Crankcase guard shall protect the crankcase, flywheel housing, accessories, and other assemblies exposed to damage from projecting obstructions likely to be encountered during normal earth moving operations. Crankcase guard shall be furnished with openings which shall provide easy access to points requiring periodic servicing and cleaning. Access openings shall be equipped with cover plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

3.10.2 Draft frame. The draft frame shall consist of the kingpin housing, gooseneck, and yoke. The frame shall support the scraper bowl and its components. Front of the frame shall be coupled to and supported by the tractor; rear shall be mounted on two power driven pneumatic tired wheels.

3.10.3 Push frame. The push frame shall incorporate a push plate. Frame shall be mounted on the drive wheels specified in 3.10.2. The frame shall support the rear power plant and power train. The push plate and frame shall withstand the forces applied to it without deformation or damage when the scraper is push-loaded with another scraper of equivalent size.

3.11 Scraper bowl. The scraper bowl shall be of the front-loading, front-lifted, open-bowl type complete with apron, blade base, ejector, cutting edges, wide bits, and locking devices. Bowl shall be free of overhead beams, structures, and hydraulic components that would obstruct loading by means of crane shovels, dragline, and loaders. Replaceable guides or wear plates shall be provided for apron, draft arm, and ejector contact surfaces. The bowl shall withstand without permanent deformation or damage:

a. Both the up force and down force of the hydraulic cylinders when it is concentrated on either end section or center section of the blade base.
b. All operating forces of the apron and ejector.
c. A load weighing not less than 75,000 pounds (lb).

When loaded as specified in c, the bowl's overall width shall not deflect more than 1/2 inch or to an extent which causes excessive rubbing, wearing, or adverse operation. Blade base shall be drilled to permit installation of the cutting edges as specified herein.

* 3.11.1 Cutting edge. Cutting edge shall consist of not less than three sections or more than four sections with an adjustable length drop center. End sections shall be identical, reversible, and interchangeable. Cutting edge sections shall be furnished with a double row of bolt holes. Bolt holes shall be spaced and countersunk in conformance to SAE J737 and SAE J740, respectively. Cutting edge cross section dimensions shall be not less than 13 inches wide and 7/8 inch thick; dimensions and tolerances shall conform to SAE J738.

3.11.2 Side bits. Side bits shall be reversible and interchangeable, and of the same material and hardness as the cutting edges.

3.11.3 Apron. Apron opening shall conform to the dimensions specified in table I. The total closing force of the apron when concentrated over a 12-inch length at the center and at either end of the lip shall not cause permanent deformation or damage to the apron. Apron shall withstand, without damage or permanent deformation, the total force of the ejector concentrated over any 80 square inches of apron area.

3.11.4 Ejector. Ejector shall completely discharge a full load of earth from the bowl and return to its loading position under positive hydraulic power. The ejector shall withstand without permanent deformation or damage the total ejection force concentrated over any 60 square inches of the ejector face.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

3.11.5 Locking devices. Locking devices shall be provided to secure the apron in a fully opened position. All locking devices shall withstand full hydraulic forces without permanent deformation. Attachment and removal of the devices shall be accomplished without use of tools. Space shall be provided on the scraper for securely storing the locking devices.

3.12 Axles. Scraper shall be equipped with heavy duty, full floating planetary drive axles. Rear axle shall be furnished with a limited slip differential. Front axle shall be provided with either a manually controlled or automatically controlled limited slip differential.

3.13 Brakes. The scraper shall be equipped with service brakes, parking brakes, and an emergency stopping system that shall conform to the minimum performance requirements of SAE J319. When brakes are air actuated, the air reservoir volume shall conform to SAE J813. Brake drum guards shall be furnished.

3.14 Wheels, rims, and tires. Wheels, rims, and tires shall be demountable as an assembly and interchangeable front to rear. Tires shall be wide base, tubeless, with rock type tread. Tire and rim sizes, ply rating and load rating at 30 mph shall conform to T&RA recommendations. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground with scraper loaded to rated gvw (6.4.3).

3.15 Power trains. Engine power shall be transmitted to the drive wheels via a torque converter coupled to a hydraulically power shifted transmission, differential, and final drive. The scraper transmissions shall be controlled and shifted together by a single lever. Torque converter oil coolers shall be furnished. Torque converter oil temperature indicators located at the operator's station shall be provided.

* 3.15.1 Scraper transmission. Each scraper transmission shall be semi-automatic or fully automatic and shall have not less than three speeds forward and one speed reverse.

3.16 Power plants. Each scraper engine shall be a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than six cylinders. The net horsepower (hp) for each engine shall be as specified herein. Each engine shall be equipped with the following:

a. Two-stage, dry type, intake air cleaner with service indicator and automatic dust ejector.
b. Full flow type oil filter.
c. Centrifugal type governor, gear driven from the engine, with governed speed setting control throughout engine load range while machine is in operation.
d. Hour meter.
e. Safety control device of the visual or audible type for high coolant temperature and low lubricating oil pressure.
f. The operator's station shall be equipped with, but not necessarily limited to, the following instruments and controls:

    1. Lubricating oil pressure indicators.
    2. Cooling liquid temperature indicators.
    3. Tachometer, (front engine only).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

4. Battery-alternator or generator indicator.
5. Start - stop control.
6. Throttle control.

g. Hood.

* 3.16.1 Scraper engines. The combined net hp rating at the flywheel of the scraper's engines shall be not less than 700 when determined as specified in SAE J816, and at a maximum governed engine speed of not more than 2200 revolutions per minute. Combined maximum torque of the scraper's engines shall be not less than 2,000 foot pounds. The total gross vehicle weight to flywheel horsepower ratio for scraper shall not exceed 270.

3.17 Power plant systems.

* 3.17.1 Starting. Both engines shall start within 5 minutes and be ready for full load operation within 15 minutes in any ambient temperature from +125°F to -20°F. The scraper shall have a 24-volt direct current cranking system. Starting aids may be either electric glow plug or ether primer. When an ether priming system is furnished, it shall be of the measured shot type with a storage capacity of not less than 12 fluid ounces.

3.17.2 Fuel. Total fuel tank capacity for the scraper shall be not less than 200 gallons. The filler-neck opening shall be not less than 3 inches in diameter. Cap shall incorporate a filter to prevent dust entering the tank when tank is vented through the cap. The cap shall be secured to the tank with a captive chain. A fuel gage located at the tank or at the operator's station shall be furnished. Two fuel tanks may be furnished.

3.17.3 Exhaust. The engines shall be equipped with exhaust mufflers and stacks. Exhaust stack outlets shall be provided with rain caps or 90° bends.

3.17.4 Cooling. Each scraper engine shall be furnished with heavy-duty cooling systems. The systems shall maintain coolant at a temperature below the boiling point when the unit is loaded to rated gvw and operating at 10,000 feet above sea level, or in an ambient air temperature of not less than 125°F. Radiator cores shall be protected against damage by heavy-duty guards. Guards shall provide ready access for cleaning and removing debris from front and around the radiators.

3.18 Electrical systems.

* 3.18.1 Ignition and starting system. The scraper shall have a 24-volt electrical system. Not less than two batteries shall be furnished. Each battery shall be 12-volt potential. The total reserve capacity rating of the batteries shall be not less than 320 minutes and the total cold cranking rating of the batteries at °F shall be not less than 1200 amperes, measured in accordance with SAE J537. Scraper shall be provided with a battery box painted with acid-resistant paint. Scraper shall be equipped with a generator or alternator having not less than a 35-ampere rating. All wiring shall be protected against wearing and weathering.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

3.18.2 Lighting system. The lighting system shall include the following:

a. Two headlamps.
b. One rearward directed floodlamp.
c. One scraper floodlamp directed to illuminate the bowl, apron, and cutting edge.
d. Instrument panel lamps to illuminate all instruments.

Lamps a, b, and c shall conform to SAE J598 and the dimensional specifications of SAE J572. Lamp circuits a and d shall be connected for actuation by a single control. Independent controls shall be provided for circuits b and c.

3.19 Hydraulic system. A hydraulic system shall be utilized for transmitting engine power to the mechanisms that operate the apron, bowl, ejector, and steering. The hydraulic system shall be complete with all operating accessories, including pump(s), motor(s), reservoir(s), valves, filters, hoses, piping, fittings, and pressure relief valve(s). Pump(s) shall supply a continuous flow of hydraulic fluid, and shall have sufficient capacity to energize all hydraulically powered components. System shall use hydraulic oil with the characteristics to withstand a temperature range from 10°F to 325°F. Pump(s) shall supply a continuous flow of hydraulic fluid, and shall have sufficient capacity to energize all hydraulically powered components. The system, including hoses and fittings, shall be capable of withstanding a bursting pressure of four times the working pressure and a maximum proof pressure of at least two times the working pressure. Hoses shall be protected against chafing through utilization of supports and protective devices. Hydraulic system shall be protected against the entrance of dirt or other foreign material. A wiper shall be provided on each hydraulic cylinder adjacent to the piston rod seal to prevent dirt and abrasives from damaging the seal or entering the cylinder. System shall be furnished with full flow filter(s) having a micron rating not greater than 25. Reservoir(s) shall have sufficient capacity to allow for maximum displacement of all cylinders at the same time, and to aid heat dissipation. Return line shall be terminated below the minimum oil level of the reservoir. Suction line shall be above the reservoir bottom and shall be protected with a readily accessible strainer. Drain plug(s) shall be of permanent magnetic type. Three separate control valves shall be provided to permit independent operation of the apron, ejector, and bowl. The apron valve shall provide control for power raise, hold, power lower, and float. The bowl valve shall permit control for power raise, power lower, and quick drop. The ejector valve shall allow control for power eject, hold, detented return, and automatic disconnect on return.

* 3.20 Steering. The scraper shall be furnished with a full hydraulic steering system. Emergency steering capability in the event of an engine or steering power source failure shall be provided in accordance with SAE J53.

3.21 Operator's station. Operator's station shall consist essentially of a seat, steering wheel, horn, safety glass wind shield with power operated windshield wiper(s), instrument panel, operating controls, and roll-over protective structure. The seat and backrest shall be covered with a durable waterproof covering. Seat and backrest dimensions and range of adjustments shall conform to SAE J899. A seat belt conforming to SAE J386 shall be provided. All controls and instruments necessary to operate the unit shall be located at the operator's station. Controls and instruments shall be clearly

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

identified for their intended operation and function. Instruments shall be readily visible to the operator when seated in a normal operating position. The roll-over protective structure shall conform to the minimum requirements of SAE J1040.

* 3.21.1 Cab. When specified (see 6.1), a cab, fully enclosed, shall be provided. The cab shall be the manufacturer's current standard production type, enclosing the operator's seat and all controls. The cab shall not obstruct the operator's view while the operator is seated. The cab shall comply with 29 CFR 1926.1001 (ROPS), and glass shall conform to 29 CFR 1926.600. The cab shall be equipped with not less than one door and one opposite side window, or one half door and one opposite side window. The door, half door, or side windows shall be provided with a lock or latch to secure the door or window in the open position. A personnel heater, defroster fan, front windshield wiper, and a rear view mirror, mounted outside and on the left hand side of the cab, shall be furnished.

3.22 Toolbox. Scraper shall be provided with a metal toolbox. The toolbox shall have a hinged lid which shall open not less than 90° and a latching device shall be provided. Toolbox shall be securely fastened to the scraper in a protected and accessible location.

* 3.23 Lifting and tying down attachments. Scraper shall be equipped with lifting and tying down attachments. Attachments shall conform to MIL-STD-209, type II or type III. Attachments shall be indicated by a data plate, except tying down attachments may be identified by stenciling or other suitable marking on the scraper. Tying down marking shall clearly indicate that the attachments are intended for tying down of the scraper on the carrier when shipped. Data plates shall be of a material which will be durable, corrosion resistant, and remain legible for the life of the equipment, and shall be mounted in a conspicuous location. The backing of the plate and the surface to which the plate is applied shall be made corrosion resistant by application of a paint or preservative compound.

3.24 Lubrication. Lubrication means shall be provided for all parts of the scraper normally requiring lubrication. Where the use of high lubrication pressure will damage grease seals, fittings with pressure release shall be used.

3.25 Identification plates. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates. Nomenclature shall be "SCRAPER, EARTHMOVING, 21 CU YD, DED".

* 3.26 Instruction and transportation plates. Instruction and transportation plates shall be of a material which will be durable, corrosion resistant, and remain legible for the life of the equipment, and shall be mounted in a conspicuous location. The backing of the plate and the surface to which the plate is applied shall be made corrosion resistant by application of a paint or preservative compound. Instruction plates shall contain clear concise instructions for the operation of all controls, and shall describe the

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

procedure to be followed in servicing the scraper. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the scraper showing the center of gravity shall be provided on the transportation plate.

3.27 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding and other interfering substances. As soon as practicable, after cleaning and before any corrosion product or other coating interfering material can result, the surface shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. The color of the finish coat shall be as specified (see 6.1).

3.28 Workmanship.

3.28.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which could be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.28.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.28.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.28.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination. Each scraper shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.2.1 First article inspection. First article inspection shall be performed on one scraper when a first article sample is required (see 3.2). This inspection shall include the examination of 4.2 and the tests of 4.3. The first article may be a standard production item from the contractor's current inventory provided the scraper meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining scrapers to be furnished under the contract.

4.3 First article tests. The first article shall be subjected to the tests specified in 4.3.1 through 4.3.4. Failure to pass one or more tests shall be cause for rejection.

4.3.1 Performance test. Scraper shall load, haul, and dump undisturbed or recompacted soil for not less than 8 hours or more than 10 hours. Each load shall be not less than the SAE J741 struck capacity rating of the bowl, and not less than one-half of the loads shall be maximum heaped. Scraper shall be push-loaded 15 times to maximum heaped capacity. Scraper shall make not less than 70 load-haul-dump cycles during this test. Haul distance shall be not less than 1,500 feet. Hauling and traveling shall be at a maximum safe speed. With a heaped payload, scraper shall negotiate a well compacted earth grade having a minimum slope of 20 percent. Length of the grade shall be not less than twice the length of the scraper. Scraper shall be driven up the grade not less than 10 times and down 10 times. Without payload, scraper shall traverse longitudinally along a grade having a minimum slope of not less than 30 percent. Grade shall be of compacted earth having a minimum length of not less than twice the scraper's length. Scraper shall make not less than two passes: one from right to left and one from left to right. During this test, scraper shall be observed to deterine conformance to:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

a. The cutting depth and width, spreading depth, and nonstop turning width specified in table I.
b. Performance requirements specified in 3.5, 3.8, 3.10.3, 3.17 and 3.19.

4.3.2 Scraper bowl test. The bowl width(s) shall be measured at 1 foot intervals both horizontally and vertically. Bowl shall be loaded to a capacity of not less than 75,000 lb. Load shall be ejected and the bowl measured again to verify conformance to 3.11. To ascertain conformance to the requirements of 3.11, 3.11.3, 3.11.4, and 3.11.5, the bowl, apron, ejector, and locking devices shall be subjected to the forces specified herein.

4.3.3 Brake test. To verify compliance with the requirements of 3.13, the brake system shall be tested in accordance with SAE J319.

4.3.4 Lifting and tying down attachment test. Lifting and tying down attachments shall be tested to determine conformance to 3.23.

4.3.5 Production sample. Upon acceptance of the first article, the first article shall remain at the manufacturing facility as a production sample, and shall be the last scraper delivered on the contract. The first article shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

4.4 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of Section 4 of MIL-T-3351. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.1), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.1).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-T-3351 with the level of preservation and the level of packing as specified (see 6.1).

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When specifications and standards shall be other than as specified (see 2.1.1).
c. When a first article scraper is to be furnished for inspection and approval (see 3.2, 4.2.1, and 6.3).
d. When cab is required (see 3.21.1).
e. Color of finish paint (see 3.27).
f. When a preproduction pack inspection is required and timeframe for submission (see 4.4).
g. Level of preservation and level of packing required (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

6.2 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete scraper or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

6.4 Definitions.

6.4.1 Recompacted soil. Soil that has been recompacted to not less than 75 percent Standard Proctor and within ±2 percent optimum moisture.

6.4.2 Maintained earth road. A maintained earth road has a smooth, firm earth surface that undulates or slightly flexes under load, and is watered and maintained regularly.

6.4.3 Gross vehicle weight. Gross vehicle weight shall consist of the weight of the scraper with a full complement of fuel, lubricants, and coolants, and a 175-lb operator, and the minimum payload as specified in table I.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Navy - YD

Preparing activity:

Navy - YD

Project No. 3805-N109

Source: https://assist.dla.mil/ Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. 3805-N109

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-28618B(YD) SCRAPER, EARTHMOVING, 21 CUBIC YARD

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used. S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:26Z
Check the source to verify that this is the current version before use.